Multilingual Tokenization

Case study deliverable: low-resource tokenization analysis on MASSIVE intent classification

Research Question

How do word-level and subword tokenizers behave across typologically different languages, especially when only a small training budget is available?

Hypothesis: subword tokenization should reduce OOV failures and improve downstream intent classification in morphologically rich or lower-resource settings.

Experimental Setup

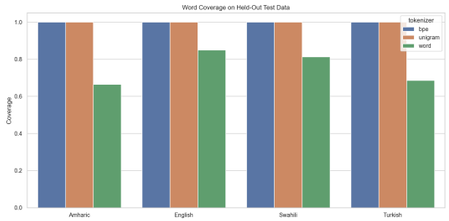
- Dataset: MASSIVE intent classification with aligned labels across languages.
- Languages: English, Turkish, Swahili, and Amharic.
- Low-resource simulation: 3,000 train examples per language.
- Tokenizers: fixed word vocabulary, SentencePiece BPE, SentencePiece unigram.
- Evaluation: intrinsic coverage metrics plus TF-IDF + Linear SVM macro-F1.

Dataset Snapshot

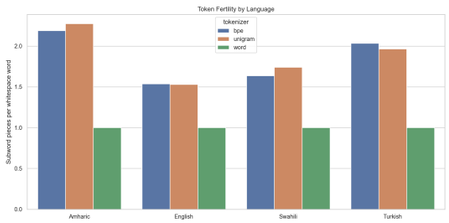
Lang	Train	Test	Unique words
Amharic	3000	2974	4578
English	3000	2974	2637
Swahili	3000	2974	3091
Turkish	3000	2974	4376

Turkish and Amharic have the largest unique word counts in the sampled training split, making them the strongest stress test for token coverage with a small vocabulary.

Intrinsic Coverage



Token Fertility



Methodological Readout

- Word coverage estimates how many held-out whitespace words can be represented without ``.
- Fertility measures how many subword pieces are needed per whitespace word.
- A good tokenizer balances broad coverage with compact segmentation and better downstream signal.

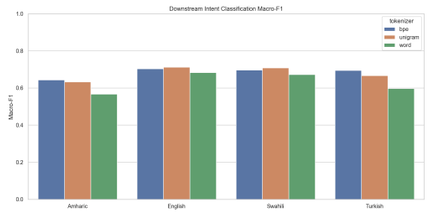
Main Findings

- Subword tokenization reached full held-out word coverage in all four languages.
- Turkish saw the largest gain over word-level tokenization: +0.098 macro-F1.
- Amharic also improved strongly with subwords: +0.076 macro-F1 over the word baseline.
- The best tokenizer is language-dependent: unigram led on English and Swahili, while BPE led on Turkish and Amharic.

Best Tokenizer by Language

Language	Best	Macro-F1
Amharic	BPE	0.643
English	UNIGRAM	0.712
Swahili	UNIGRAM	0.709
Turkish	BPE	0.695

Downstream Performance



Macro-F1 confirms that coverage gains usually transfer to the intent classification task, especially in Turkish and Amharic.

Conclusion

The experiments support the assignment hypothesis: word-level tokenization breaks down faster in lower-resource and morphologically rich languages, while subword tokenization consistently improves coverage and usually improves downstream intent classification. For multilingual low-resource pipelines, BPE and unigram are both safer default choices than a fixed word vocabulary.